PROJECT REPORT

FOR

Foundations of Artificial Intelligence

AI-Based Loan Eligibility Assessment System

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report "**AI-Based Loan Eligibility Assessment System using Forward Chaining Algorithm**" is the bonafide work of " Natarajan S (240701347) & Muhilan C (240701338) " who carried out the project work under my supervision.

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ABSTRACT:

The **AI-Based Loan Eligibility Assessment System using Forward Chaining Algorithm** is an intelligent, rule-based loan assessment platform designed to provide transparent, consistent, and explainable loan eligibility decisions. The system combines a rule-based expert system, Forward Chaining inference, Agentic AI workflow, and Explainable AI to automate the assessment process.

Applicants provide details such as income, employment information, existing financial obligations, loan amount, and demo credit score. The system validates the submitted information and derives additional financial facts such as EMI and Debt-to-Income (DTI) ratio. These facts are then processed by a Forward Chaining inference engine, which evaluates a modular knowledge base containing predefined loan eligibility rules. Based on the satisfied rules, the system produces one of three outcomes: **ELIGIBLE, NOT_ELIGIBLE, or MANUAL_REVIEW**.

The Agentic AI workflow assists with information validation, missing-data handling, fact generation, and explanation generation, while the final eligibility decision remains exclusively with the Forward Chaining engine. The system also records the rules triggered during assessment, enabling users and administrators to understand the reasons behind each decision. Additional features include EMI calculation, What-If analysis, loan amount recommendation, risk analysis, assessment history, PDF reports, and an administrative dashboard for managing applications and knowledge-base rules.

The proposed system demonstrates how Artificial Intelligence and knowledge-based reasoning can be applied to financial decision-support systems while maintaining transparency and explainability. The rules and thresholds used in the project are intended only for academic demonstration and do not represent actual banking or credit-underwriting criteria.

1. INTRODUCTION

Loan eligibility assessment is an important process in the financial sector, where an applicant's financial information is evaluated to determine whether they satisfy predefined lending conditions. Traditional loan assessment processes may involve multiple factors such as income, credit score, existing financial commitments, employment status, loan amount, and repayment capacity. As the number of applications increases, there is a need for systems that can process these factors consistently while also providing clear explanations for their decisions.

The **AI-Based Loan Eligibility Assessment System using Forward Chaining Algorithm** is designed as an intelligent decision-support system that automates the loan assessment process using **Artificial Intelligence and rule-based reasoning**. The system collects applicant information, validates the inputs, calculates financial indicators such as **EMI and Debt-to-Income (DTI) ratio**, and processes the resulting facts using a predefined knowledge base of loan assessment rules.

At the core of the system is the **Forward Chaining inference engine**, which evaluates available facts against the conditions of the rules and derives new facts until a final decision is reached. The system produces three possible outcomes: **ELIGIBLE, NOT_ELIGIBLE, or MANUAL_REVIEW**. Each decision is accompanied by the rules that were triggered, allowing users to understand how the result was obtained.

The project also incorporates an **Agentic AI workflow** to coordinate tasks such as checking missing information, validating inputs, generating financial facts, invoking the Forward Chaining engine, and preparing explanations. However, the agent does not make or modify the eligibility decision. This separation ensures that the **Forward Chaining engine remains the authoritative decision-making component**.

In addition, the system provides features such as **EMI calculation, What-If simulation, risk analysis, loan amount recommendation, assessment history, PDF reports, and**

administrative rule management. These features demonstrate how AI-based reasoning can be integrated with a modern web application to create a transparent and explainable loan assessment platform.

The project is intended for **academic and demonstration purposes**. The rules, credit-score ranges, and financial thresholds used in the system are sample values and **do not represent actual banking or credit-underwriting criteria**.

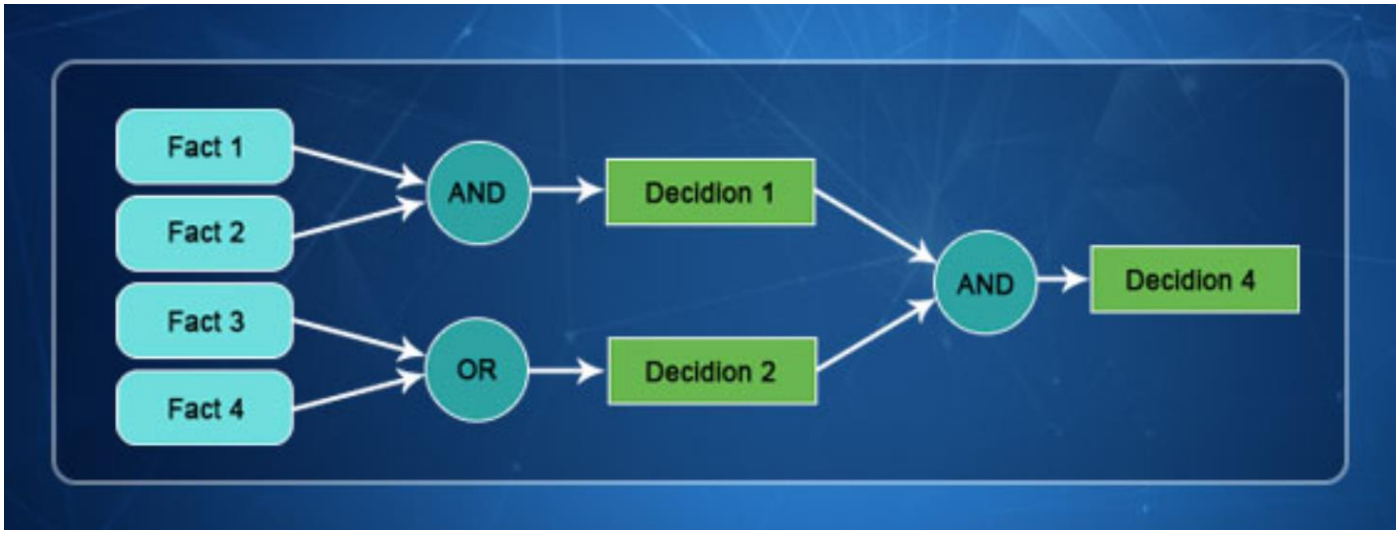


Fig 1.1 Forward Chaining Algorithm

2. LITERATURE REVIEW

Rule-Based Expert Systems for Loan Assessment:

Jackson and Smith (2020) studied the application of rule-based expert systems in financial decision-making and found that predefined rules can be used to evaluate financial conditions consistently. Their research highlighted the advantage of transparent decision-making, where the conditions leading to a particular result can be clearly identified. This concept influenced the development of the rule-based knowledge base in the proposed **AI-Based Loan Eligibility Assessment System**.

Kumar et al. (2021) investigated automated loan assessment systems and observed that factors such as income, credit score, existing liabilities, and repayment capacity can be represented as structured rules for eligibility evaluation. The study emphasized that automated assessment can reduce manual processing and provide consistent results. This finding influenced the use of structured financial facts and modular eligibility rules in the proposed system.

Forward Chaining and Knowledge-Based Reasoning:

Patel and Verma (2020) studied forward chaining as a data-driven inference technique and found that it can derive new conclusions by repeatedly applying rules to known facts. Their research demonstrated the usefulness of forward chaining in decision-support and expert systems. This concept forms the core of the proposed system, where applicant facts are processed through a sequence of predefined rules to determine the final loan eligibility outcome.

Singh et al. (2022) examined rule-based decision-making systems and highlighted the importance of recording the rules activated during the inference process. The study suggested that tracking the reasoning path can improve transparency and make automated

decisions easier to understand. This approach influenced the implementation of **fired-rule tracking and explanation generation** in the proposed loan assessment system.

Explainable Artificial Intelligence in Financial Decision-Making:

Ribeiro et al. (2016) introduced LIME, a technique for explaining predictions made by machine-learning models. Their work demonstrated the importance of providing understandable explanations for automated decisions, particularly when users need to understand why a particular prediction was produced. This concept motivated the inclusion of an explainability component in the proposed system.

Arrieta et al. (2020) reviewed Explainable Artificial Intelligence techniques and emphasized the importance of transparency, interpretability, and human understanding in AI systems. Their research highlighted the need for AI systems to provide meaningful information about how decisions are generated. This influenced the proposed system's approach of displaying the **rules fired, reasons, and associated risk factors** along with the eligibility result.

3. PROPOSED SYSTEM

The proposed system is an **AI-Based Loan Eligibility Assessment System using Forward Chaining Algorithm** that provides an automated, transparent, and explainable approach to evaluating loan applications. The system combines a **rule-based expert system, Forward Chaining inference engine, Agentic AI workflow, and Explainable AI** within a web-based application.

The system accepts applicant information such as income, employment details, existing financial obligations, loan amount, and demo credit score. The submitted information is first validated, and important financial indicators such as **EMI and Debt-to-Income (DTI) ratio** are calculated. These values are converted into structured facts and supplied to the Forward Chaining engine.

The **Forward Chaining engine acts as the authoritative decision-making component**. It compares the available facts against a modular knowledge base containing predefined loan assessment rules. When the conditions of a rule are satisfied, the rule is fired and its conclusion is added to the working memory. This process continues until no additional facts can be derived. Based on the final derived decision facts, the system generates one of three outcomes: **ELIGIBLE, NOT_ELIGIBLE, or MANUAL_REVIEW**.

An **Agentic AI workflow** is used to coordinate the overall assessment process. It checks for missing information, validates inputs, performs financial calculations, invokes the Forward Chaining engine, and prepares explanations and risk factors. The agent does **not** make or override the eligibility decision, ensuring that the final result is determined only by the rule-based inference engine.

The system also provides an **Explainable AI panel** that displays the rules triggered during assessment and explains the factors contributing to the result. Users can perform **What-If simulations** by changing parameters such as loan amount or financial

conditions and compare the resulting decision and rule differences.

Major Components

1. **User Authentication and Authorization**
    - Registration and login using JWT authentication.
    - Separate USER and ADMIN roles.
    - Backend-enforced role-based access control.
2. **Loan Application Module**
    - Multi-step application form.
    - Collection and validation of applicant and loan information.
    - Storage of submitted applications in MongoDB.
3. **Financial Fact Generator**
    - Calculates EMI.
    - Calculates DTI ratio.
    - Generates structured facts required by the inference engine.
4. **Knowledge Base**
    - Contains modular loan assessment rules.
    - Supports multiple conditions and priorities.
    - Rules can be managed by administrators.
5. **Forward Chaining Engine**
    - Loads applicant facts.
    - Evaluates rules sequentially.
    - Fires satisfied rules.
    - Derives new facts.
    - Produces the final eligibility decision.
    - Records all fired rules.
6. **Agentic AI Workflow**
    - Checks missing information.
    - Validates applicant data.

- Coordinates financial calculations.
- Invokes the Forward Chaining engine.
- Generates explanations and suggestions.
- Stores the completed assessment.

7. **Explainable AI Module**
    - Displays triggered rules.
    - Provides reasons for the decision.
    - Identifies relevant risk factors.
    - Generates downloadable assessment reports.
8. **What-If Simulator**
    - Allows users to modify selected application values.
    - Re-runs the rule-based assessment.
    - Displays before-and-after results.
    - Shows differences in triggered rules.
9. **Admin Module**
    - Application and user management.
    - Assessment review.
    - Knowledge-base rule CRUD operations.
    - Analytics and application statistics.

3.1 ARCHITECTURE DIAGRAM

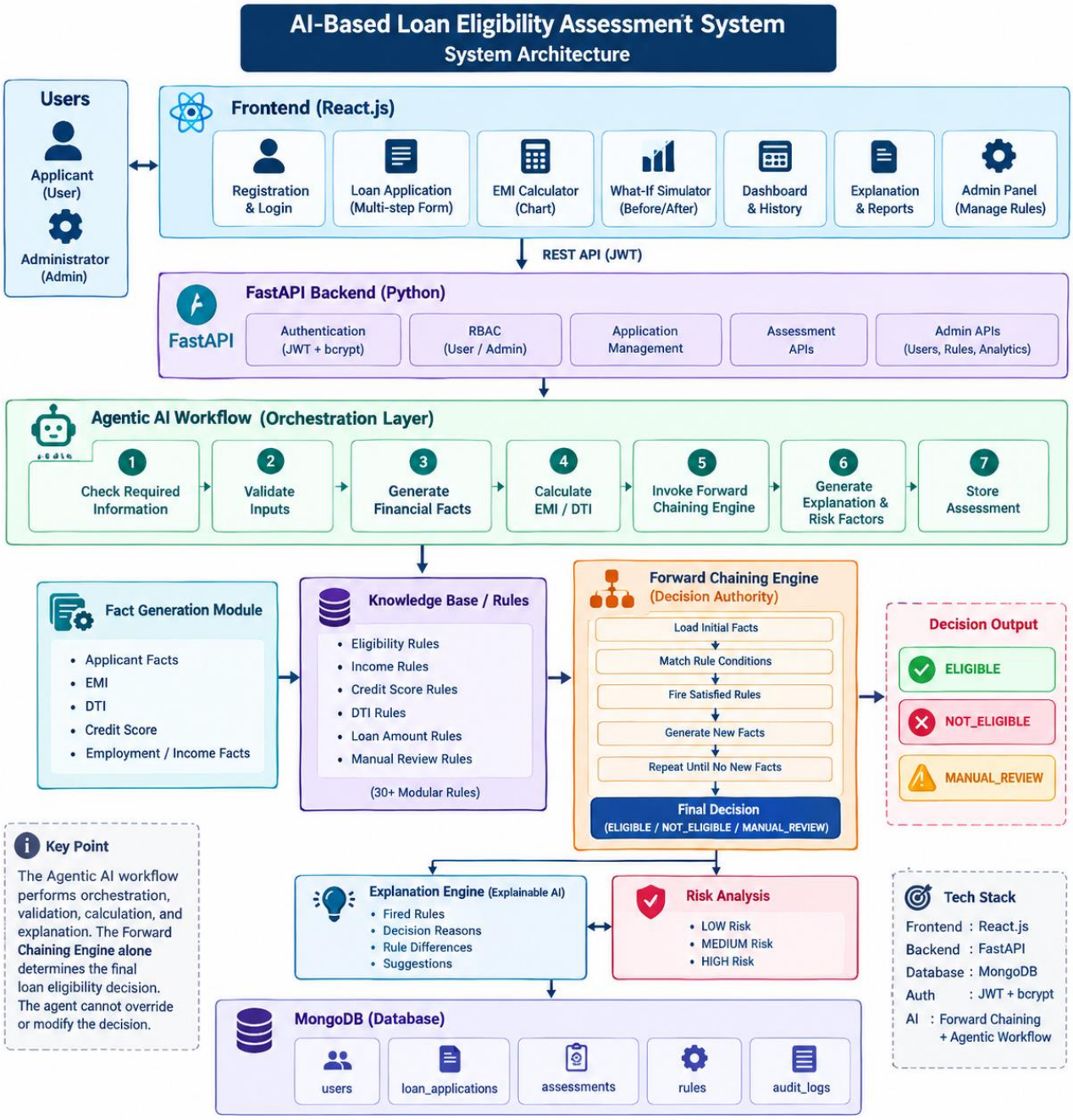


Figure 3.1: System Architecture of DuckPersona

4. MODULES DESCRIPTION

The AI-Based Loan Eligibility Assessment System is organized into two major modules that work together to provide an intelligent, transparent, and explainable loan assessment environment. The first module manages user interaction, authentication, loan applications, calculations, and result visualization, while the second module manages the AI assessment workflow, Forward Chaining inference, rule processing, explanation generation, and backend operations. The integration of these modules allows users to submit loan applications, receive automated eligibility decisions, understand the reasons behind those decisions, and perform What-If analysis.

4.1 MODULE 1 – USER INTERFACE AND LOAN APPLICATION MODULE

The User Interface and Loan Application Module forms the primary interaction layer of the system. It provides a simple and interactive web interface where users can register, log in, submit loan applications, calculate EMI, view assessment results, and analyze different financial scenarios. The module is developed using React.js and communicates with the backend through REST APIs.

The module provides a multi-step loan application form where users enter information such as personal details, employment status, income, existing financial obligations, loan amount, loan tenure, and demo credit score. Input validation is performed to ensure that the submitted information satisfies the required format and constraints.

An EMI calculator is provided to help users estimate their monthly repayment amount. The module also includes an assessment dashboard where users can view the final eligibility result, fired rules, risk factors, and explanations generated by the system.

The What-If Simulator allows users to modify selected loan parameters and compare the original assessment with the modified assessment. The module displays the before-and-after decisions and the rules that changed during the simulation.

Main functions:

- User registration and login.
- JWT-based authentication.
- Multi-step loan application.
- Applicant information collection and validation.
- Demo credit score input.
- EMI calculation and visualization.
- Loan eligibility result display.
- What-If simulation.
- Assessment history.
- Explanation and risk-factor visualization.
- PDF assessment report access.
- Admin dashboard and rule management interface.

4.2 MODULE 2 – AI ASSESSMENT AND BACKEND MODULE

The AI Assessment and Backend Module is responsible for processing loan applications and generating the final eligibility assessment. The backend is developed using Python and FastAPI and communicates with the React frontend through REST APIs.

The module first validates the submitted application and checks whether all required information is available. The Agentic AI workflow coordinates the assessment process by generating financial facts and calculating important values such as EMI and Debt-to-Income (DTI) ratio.

The generated applicant facts are passed to the Knowledge Base, which contains modular loan assessment rules. The Forward Chaining Engine evaluates these rules against the available facts. When a rule's conditions are satisfied, the rule is fired and its conclusion

is added as a new fact. The process continues until no additional facts can be derived.

The Forward Chaining engine then produces one of three final outcomes:

- ELIGIBLE
- NOT_ELIGIBLE
- MANUAL_REVIEW

The system records every fired rule so that the decision can be explained to the user. The backend also generates rule-based risk factors and prepares the information required for assessment reports.

An important feature of this module is the separation between the Agentic AI workflow and the decision engine. The agent performs orchestration, validation, calculation, and explanation tasks, but it does not make or modify the eligibility decision. The Forward Chaining Engine alone is responsible for the final eligibility outcome.

Main functions:

- REST API request handling.
- JWT authentication and authorization.
- User and application validation.
- EMI and DTI calculation.
- Financial fact generation.
- Knowledge-base rule processing.
- Forward Chaining inference.
- Eligibility decision generation.
- Fired-rule tracking.
- Explainable decision generation.
- PDF report generation.

5. IMPLEMENTATION AND RESULTS

The AI-Based Loan Eligibility Assessment System was implemented as a web-based intelligent decision-support application that combines a modern loan application interface with a rule-based AI assessment engine. The frontend was developed using React.js, while the backend was implemented using Python and FastAPI. MongoDB was used for storing users, loan applications, assessments, rules, and audit logs.

The application connects the frontend and backend through REST API communication. JWT authentication is used to secure communication and control access based on user roles. After submitting a loan application, the backend validates the information, generates financial facts, and invokes the Forward Chaining inference engine.

The Forward Chaining engine evaluates the applicant's facts against the predefined knowledge-base rules and produces the final eligibility outcome. The system also records the rules triggered during the assessment, enabling the user to understand the reasoning behind the result.

The implemented system provides ELIGIBLE, NOT_ELIGIBLE, and MANUAL_REVIEW outcomes along with explanations and risk factors. The What-If module allows users to change selected input values and observe how the rule-based decision changes.

5.1 EXPERIMENTAL SETUP

The frontend of the system was developed using React.js, React Router, Axios, and Chart.js to provide a responsive user interface, navigation, API communication, and graphical representation of financial information.

The backend was developed using Python, FastAPI, Pydantic, JWT, and bcrypt. FastAPI provides the REST API layer, while JWT and bcrypt are used for authentication and

password security.

MongoDB is used as the database for storing user accounts, loan applications, assessments, knowledge-base rules, and audit records.

The AI component consists of a custom Forward Chaining inference engine and an Agentic AI workflow. The Forward Chaining engine evaluates the applicant facts against the predefined rules, while the Agentic workflow coordinates validation, financial calculations, inference invocation, and explanation generation.

The experimental setup includes different applicant scenarios such as:

- Applicant with a strong financial profile → ELIGIBLE
- Applicant with poor credit or high DTI → NOT_ELIGIBLE
- Borderline applicant profile → MANUAL_REVIEW
- Modified loan amount through What-If simulation → Comparison of assessment results

The system records the fired rules for each assessment, allowing the results to be verified and explained during demonstration and evaluation.

Note: The credit-score ranges, financial thresholds, and eligibility rules used in the implementation are demonstration rules for academic purposes only and do not represent actual bank underwriting criteria.

4.1 RESULTS

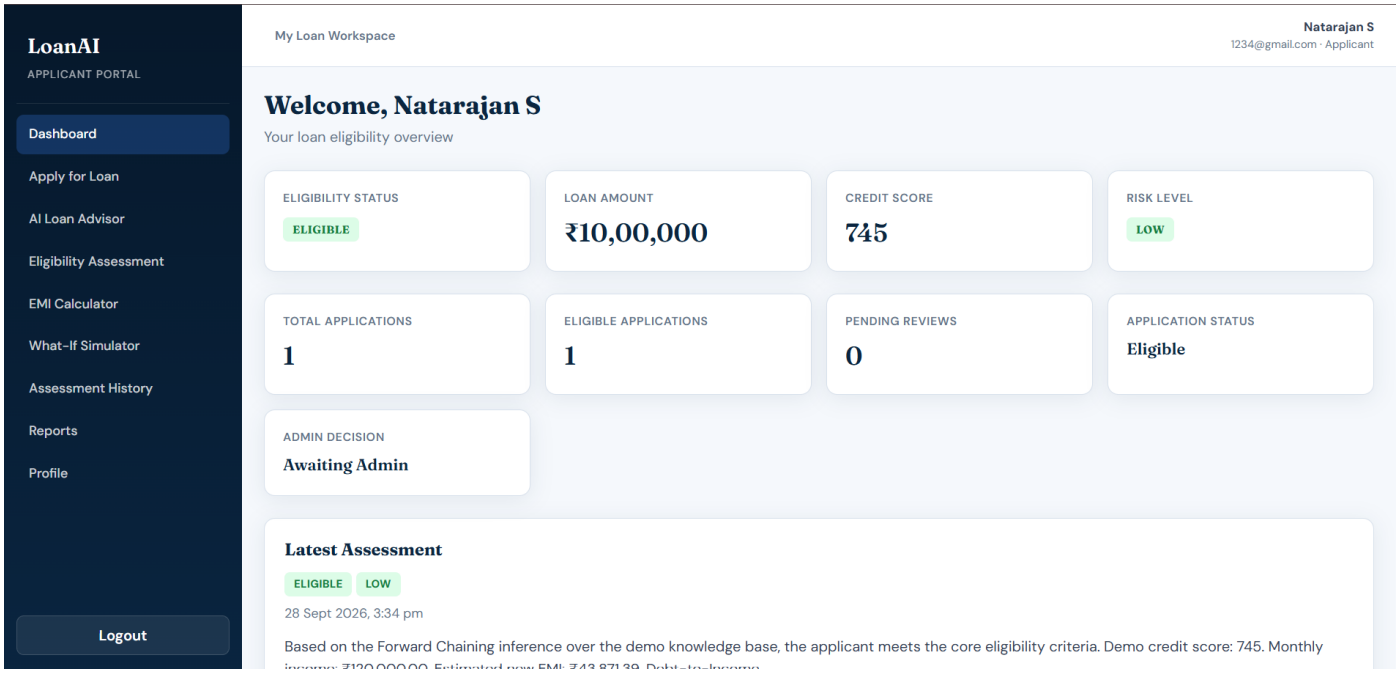


Figure 5.2.1 User Dashboard

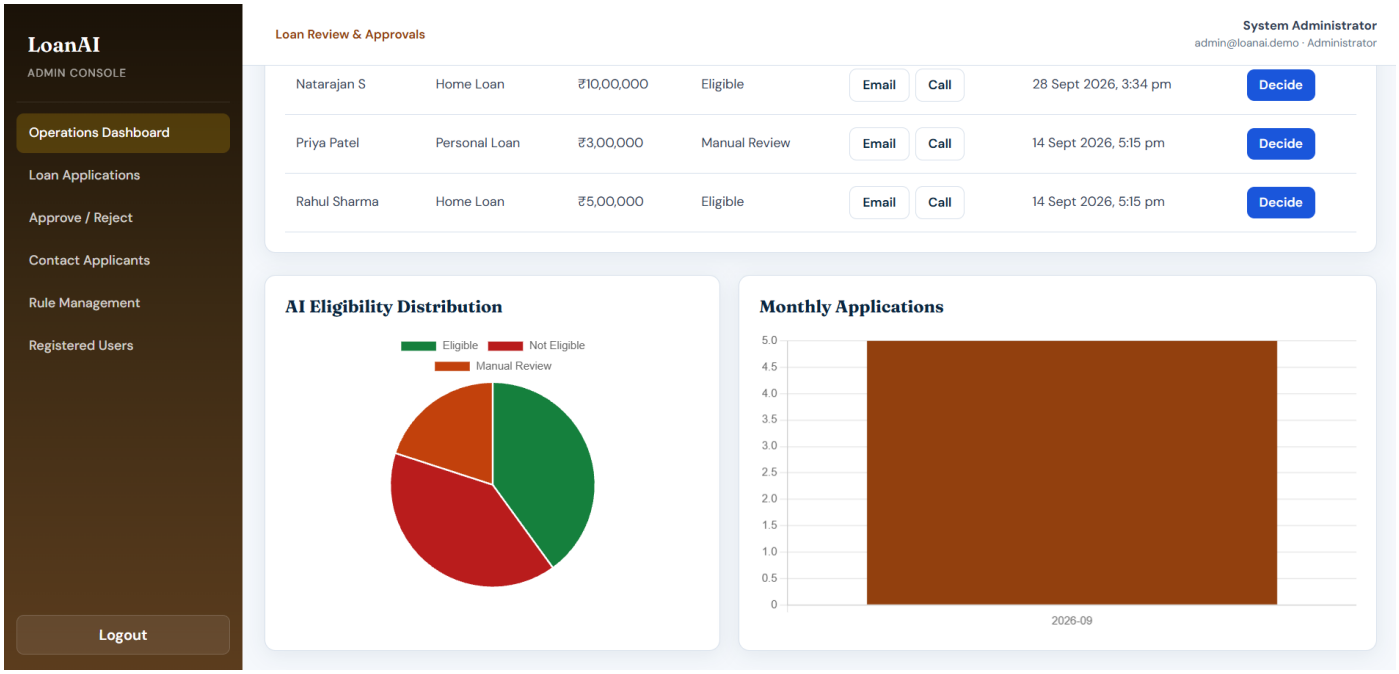


Figure 5.2.2 Admin Dashboard

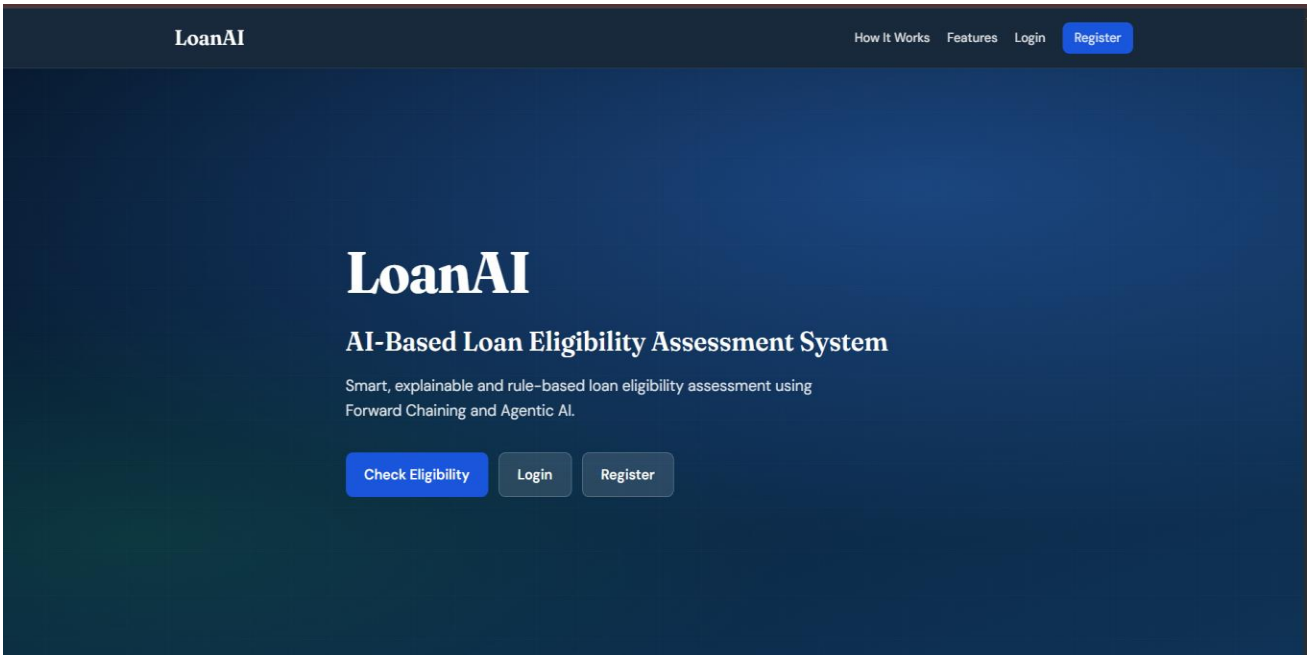


Figure 5.2.3 Login Or Registeration Page

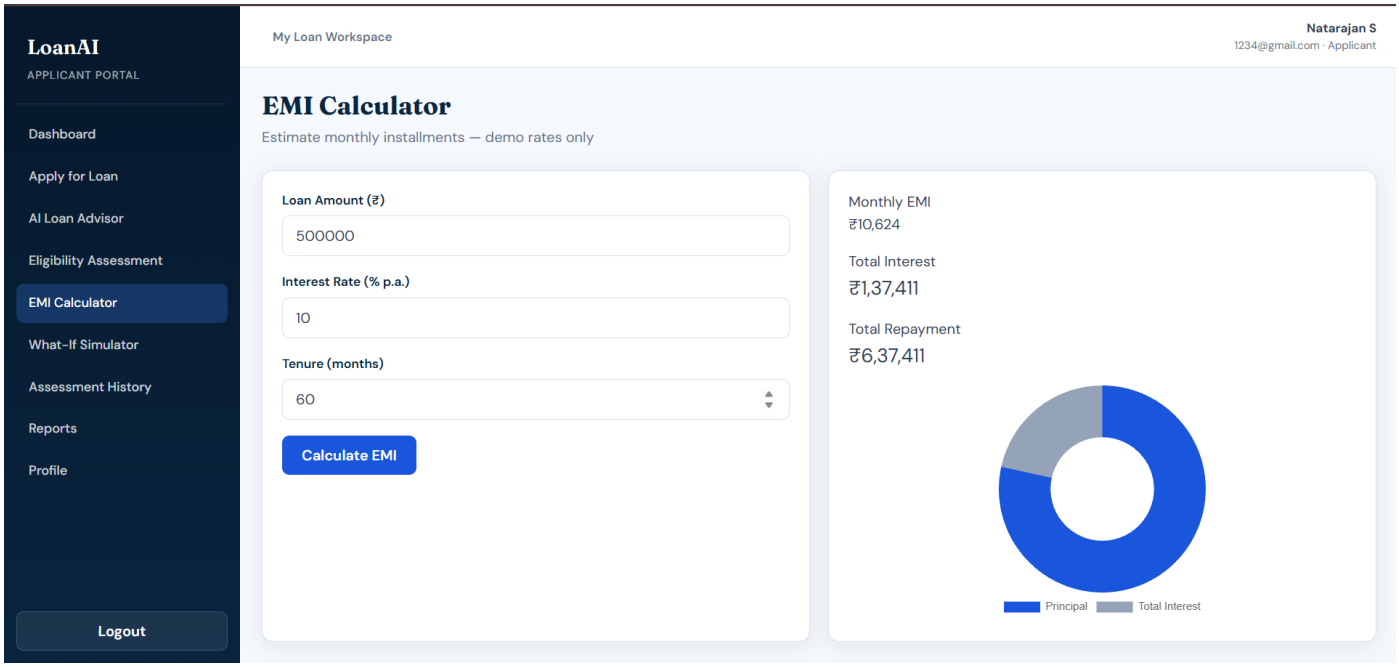


Figure 5.2.4 EMI Calculator



Fig 5.2.5 Report Creation

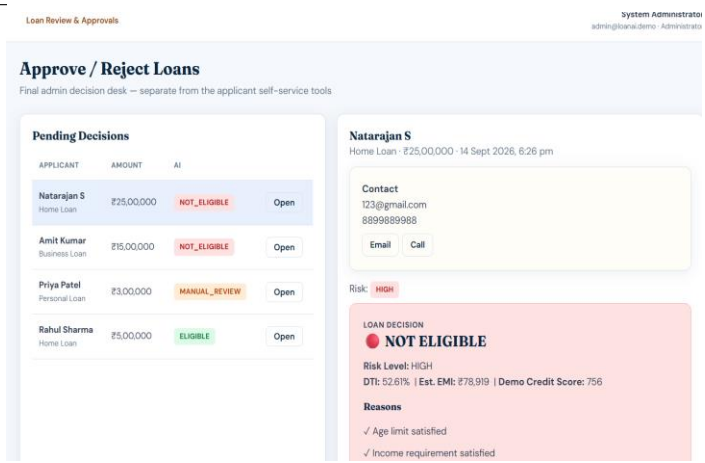


Fig 5.2.6 Approve/Reject Loan

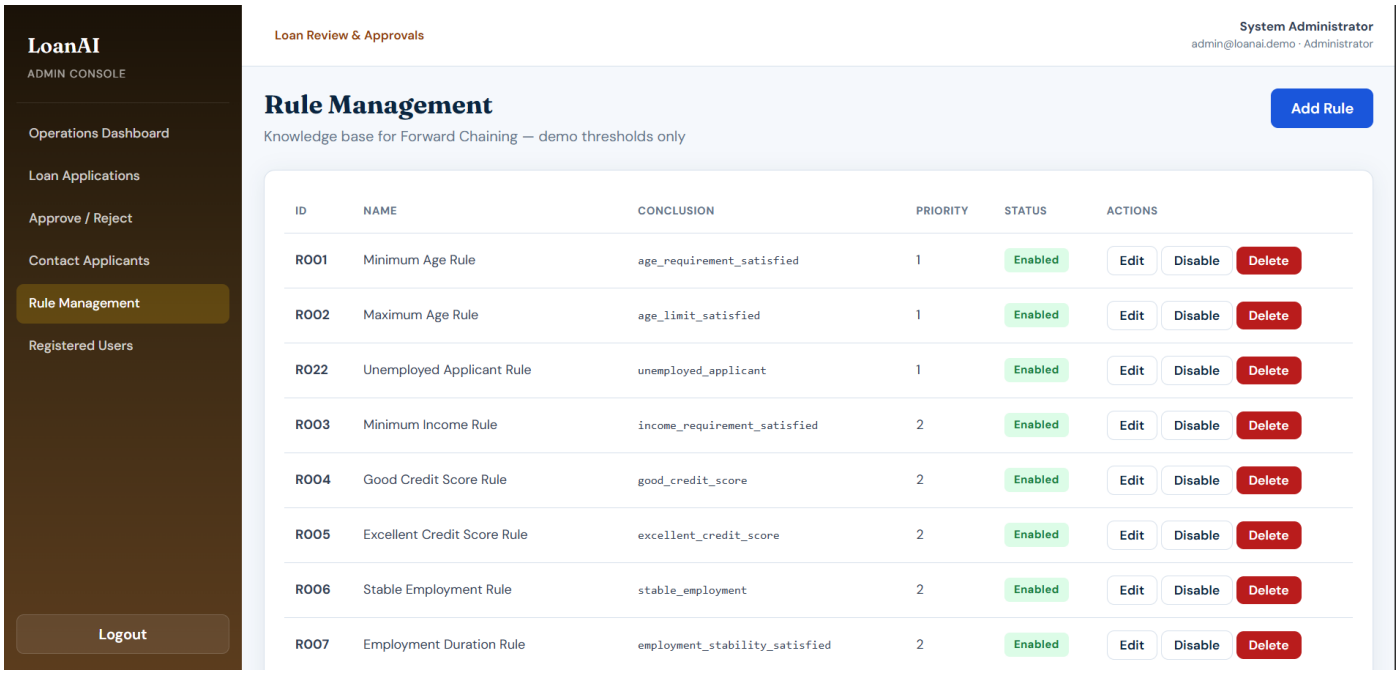


Figure 5.2.7 Rule Management

5. CONCLUSION AND FUTURE WORK

The **AI-Based Loan Eligibility Assessment System using Forward Chaining Algorithm** is an intelligent and explainable platform developed to automate the loan assessment process. The system integrates **React.js, FastAPI, MongoDB, Agentic AI workflow, a rule-based Knowledge Base, and a Forward Chaining inference engine**.

The system also provides **Explainable AI, risk analysis, What-If simulation, assessment history, PDF reports, and administrative rule management**. This approach makes the assessment process deterministic, transparent, and easy to understand. The implemented rules and thresholds are intended for academic demonstration and do not represent actual banking underwriting criteria.

FUTURE WORK

In the future, the system can be enhanced by integrating authorized credit-bureau APIs to obtain verified credit information instead of relying on user-entered demo scores. Additional features such as automated document verification, advanced fraud detection, rule versioning, conflict detection, improved analytics, and email notifications for manual-review applications can also be introduced. These enhancements can improve the functionality and reliability of the system.

The Agentic AI workflow can also be extended using advanced AI orchestration frameworks and an optional LLM-based conversational advisor for natural-language assistance. However, the Forward Chaining engine can continue to remain the authoritative decision-making component. Production-level improvements such as HTTPS, rate limiting, secure secret management, refresh-token rotation, and enhanced audit logging can further strengthen the system and support its evolution into a more comprehensive explainable financial decision-support platform.

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